

Redundancy Resolution in Motion Planning

7R robot · one supplied workspace motion, many joint motions

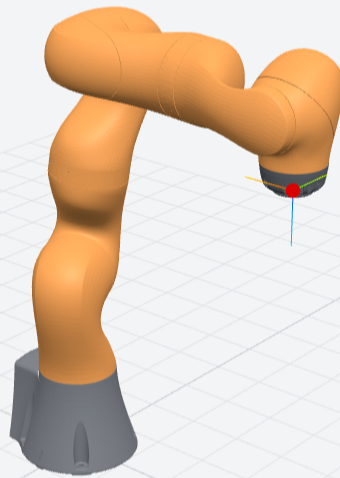
Understand the null space, compare cost functions, and explore numerical and analytical strategies on the same rectangular path.

A 7R robot can reconfigure while keeping its tool pose fixed

$$\mathbf{q}_A \neq \mathbf{q}_B, \quad f(\mathbf{q}_A) = f(\mathbf{q}_B) = {}^0T_{t,d}$$

Configuration A · $q_3 = -30^\circ$

Tool t · same position and orientation

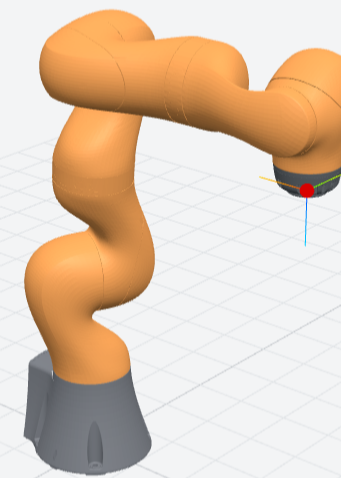


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$q_A [^\circ]$: 26.2, 8.2, -30.0, -84.4, 4.1, 88.6, -3.6

Configuration B · $q_3 = +30^\circ$

Tool t · same position and orientation



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$q_B [^\circ]$: -26.2, 8.2, 30.0, -84.4, -4.1, 88.6, 3.6

The same tool **position and orientation** admit a continuous family of nearby configurations at a regular pose. Along a supplied path ${}^0T_{t,d}(s)$, redundancy lets us choose how the arm moves internally.

Seven joint degrees of freedom, six tool-pose degrees of freedom

What is a degree of freedom?

One independent coordinate needed to specify a configuration. This serial 7R robot has seven independent joint angles; its tool pose has three position and three orientation coordinates.

$${}^0\mathcal{V}_t = \begin{bmatrix} {}^0\dot{\mathbf{p}}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = {}^0J_t(\mathbf{q})\dot{\mathbf{q}}$$

${}^0J_t \in \mathbb{R}^{6 \times 7}$. The index t denotes the tool; the superscript 0 denotes the world frame.

Kernel = null space

$$\ker J_t = \{\mathbf{v} \in \mathbb{R}^7 : J_t\mathbf{v} = \mathbf{0}\}$$

The kernel of a matrix is the set of vectors it maps to zero. For the Jacobian, these are joint velocities that produce zero instantaneous tool velocity.

One internal degree of freedom at full task rank

$$\dim \ker J_t = 7 - \text{rank } J_t = 7 - 6 = 1$$

At a regular pose, rank six means that all six tool-velocity components can be commanded locally. One independent joint-velocity direction remains for internal motion.

Every joint contributes a screw evaluated at the tool point

1 · Locate the axis

Multiply the URDF transforms along the chain to obtain a point ${}^0\mathbf{p}_i$ on joint axis i and its world-frame rotation 0R_i .

2 · Express it in the world frame

$${}^0\mathbf{a}_i = {}^0R_i {}^i\hat{\mathbf{a}}_i$$

Here frame i is attached to joint axis i . Rotate the URDF axis into world coordinates using 0R_i .

3 · Take the cross product

$${}^0J_{t,i} = \begin{bmatrix} {}^0\mathbf{a}_i \times ({}^0\mathbf{p}_t - {}^0\mathbf{p}_i) \\ {}^0\mathbf{a}_i \end{bmatrix}$$

Stack all seven columns in joint order.

$${}^0J_t = \begin{bmatrix} {}^0J_{t,1} & {}^0J_{t,2} & \cdots & {}^0J_{t,7} \end{bmatrix}$$

We use angular velocity, not Euler-angle rates. The robot models and the analytical IK are checked against the same native KUKA iiwa 7 geometry.

Different joint velocities can produce the same desired tool velocity

1 · Take two configurations and two joint velocities

Let $\mathbf{q}_A, \mathbf{q}_B$ realize the same tool pose, with respective velocities $\dot{\mathbf{q}}_A, \dot{\mathbf{q}}_B$.

$$J_t(\mathbf{q}_A)\dot{\mathbf{q}}_A = \mathcal{V}_d, \quad J_t(\mathbf{q}_B)\dot{\mathbf{q}}_B = \mathcal{V}_d$$

The Jacobians generally differ. Their velocity difference is not automatically a null vector of either Jacobian.

What is $\mathcal{V}_d$?

$$\mathcal{V}_d \equiv {}^0\mathcal{V}_{t,d} = \begin{bmatrix} {}^0\dot{\mathbf{p}}_{t,d} \\ {}^0\boldsymbol{\omega}_{t,d} \end{bmatrix}$$

The desired tool velocity: three linear and three angular components, expressed in the world frame. Holding the complete tool pose fixed means $\mathcal{V}_d = \mathbf{0}$.

2 · Fix one configuration $\mathbf{q}$

At this same configuration, take two velocity solutions $\dot{\mathbf{q}}_a, \dot{\mathbf{q}}_b$ of the same command.

$$J_t\dot{\mathbf{q}}_a = J_t\dot{\mathbf{q}}_b = \mathcal{V}_d$$

$$J_t(\dot{\mathbf{q}}_b - \dot{\mathbf{q}}_a) = \mathbf{0}$$

3 · $\mathbf{n}$ is a null vector

$$J_t\mathbf{n} = \mathbf{0}, \quad \|\mathbf{n}\| = 1, \quad \dot{\mathbf{q}} = \dot{\mathbf{q}}_p + \mathbf{n}\nu$$

$\mathbf{n}$ is a nonzero direction in the kernel (null space). $\dot{\mathbf{q}}_p$ is one velocity solution, and the scalar ν sets the signed speed along the null direction.

The pseudoinverse follows the task; the projector selects internal motion

One particular solution

$$J^+ = J^T(JJ^T)^{-1}$$

For rank six, $J^+\mathcal{V}_d$ is the solution with minimum Euclidean joint-rate norm.

Project into the null space

$$N = I - J^+J$$

$JN = 0$, $N^2 = N$, and $N^T = N$. For one redundant degree, $N = \mathbf{n}\mathbf{n}^T$ if $\|\mathbf{n}\| = 1$.

All instantaneous solutions

$$\dot{\mathbf{q}} = J^+\mathcal{V}_d + N\mathbf{z}$$

Here $J = {}^0J_t(\mathbf{q})$. The free vector $\mathbf{z}$ expresses our secondary preference.

These identities describe the exact pseudoinverse. Damped least squares improves numerical conditioning, but its associated matrix is generally not an exact null-space projector.

Change the redundant angle and watch the arm reconfigure

Hold the tool at the rectangle's centre, ${}^0\mathbf{p}_t = (0.45, 0, 0.60)$ m, with its orientation fixed. This pose has eight continuous IK branches over $q_3 = -170^\circ$ to $+170^\circ$. Follow the other six joint angles; grey portions exceed joint limits.

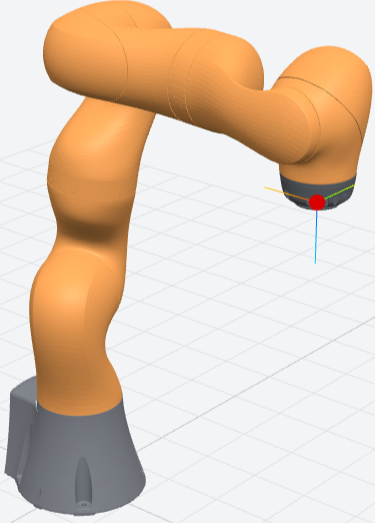
Fixed tool pose · centre

x0.45y0.00z0.60m

Rebuild curvesDefault

KUKA iiwa 7 · native joint limits

tool frame held fixed

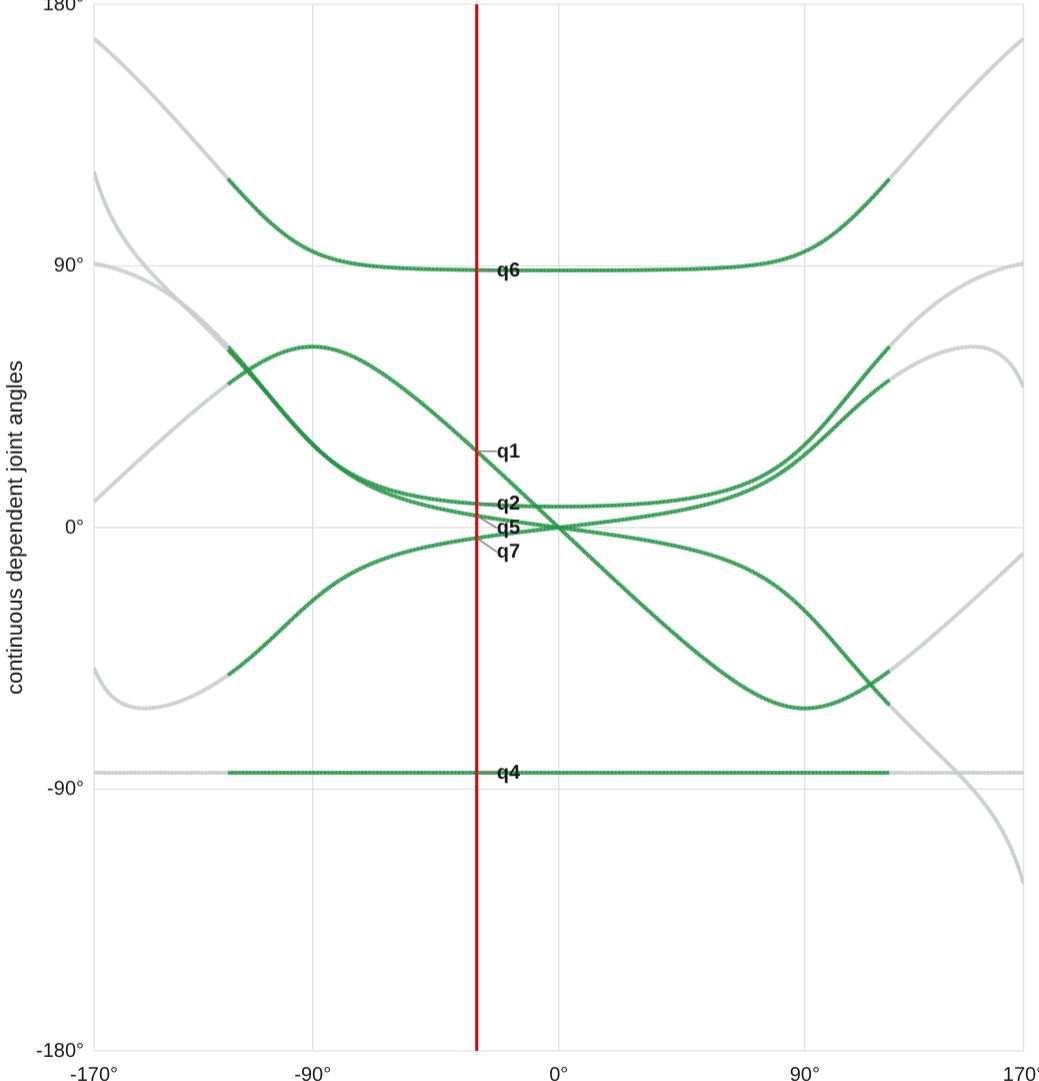


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Continuous self-motion · fixed pose

☐ All 8 IK curves

IK 5 · continuous joint angles, fixed tool pose



continuous dependent joint angles

redundant angle q_3

within joint limits

outside joint limits

selected q_3

Redundant angle

-30°

IK 1joint limit

IK 2joint limit

IK 3joint limit

IK 4joint limit

IK 5within limits

IK 6joint limit

IK 7joint limit

IK 8joint limit

$J \dot{q} = 0$

7 joint velocities · 6 tool constraints

IK 5 · $q_3 = -30.0^\circ$

tool position error $1.2e-16$ m

tool rotation error $9.0e-16$ rad

minimum joint margin 31.4°

Full J: regular · nullity 1

$q = 26, 8, -30, -84, 4, 89, -4^\circ$

Move q_3 along a continuous fixed-pose curve. Grey sections are mathematical configurations beyond the robot's joint limits.

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Minimize a cost while preserving the prescribed tool pose

Choose a preferred configuration

$$\min_{\mathbf{q}} h(\mathbf{q}) \quad \text{subject to} \quad f(\mathbf{q}) = {}^0T_{t,d}$$

h is a scalar cost: lower values mean a more desirable posture. The constraint keeps the commanded tool position and orientation. Joint limits must also be checked.

Take a local projected descent step

$$\dot{\mathbf{q}} = J^+ \mathcal{V}_d - \gamma N \nabla h, \quad \gamma \geq 0$$

$J = {}^0J_t$, $N = I - J^+ J$. The gradient ∇h points toward increasing cost; $-N \nabla h$ keeps its descent component in the null space. The gain γ sets the strength of this secondary motion.

Derive the change in cost

$$\begin{aligned} \dot{h} &= \nabla h^T \dot{\mathbf{q}} \\ &= \nabla h^T J^+ \mathcal{V}_d - \gamma \|N \nabla h\|^2 \end{aligned}$$

The first line is the chain rule. Since $N^T = N$ and $N^2 = N$, $\nabla h^T N \nabla h = \|N \nabla h\|^2$.

Fixed tool pose versus a moving tool

With $\mathcal{V}_d = \mathbf{0}$, $\dot{h} = -\gamma \|N \nabla h\|^2 \leq 0$: the cost decreases for $\gamma > 0$ whenever the projected gradient is nonzero.

With a moving tool, the primary-task term can increase the cost faster than the secondary motion reduces it.

Projected descent is a local optimization method. A zero projected gradient does not prove a global minimum, and the cost does not guarantee that joint limits can be avoided.

Choose one cost function h to express the secondary objective

Let $q_{i,c} = (q_{i,\max} + q_{i,\min})/2$, $r_i = (q_{i,\max} - q_{i,\min})/2$, and $u_i = (q_i - q_{i,c})/r_i$. Each row defines a separate choice of $h(\mathbf{q})$. The reference posture is the selected starting IK.

Interpret u_i : it is the joint's dimensionless position relative to the centre of its allowed range. $u_i = 0$ is the centre, $u_i = -1$ is the lower limit, and $u_i = 1$ is the upper limit. Thus $|u_i| > 1$ means the joint is outside its limits; this normalization lets us compare joints with different ranges.

Strategy	Alternative definitions of h	What the arm prefers
No secondary cost	$\mathbf{z} = \mathbf{0}$	Minimum joint-rate norm at each step; no explicit joint-limit avoidance.
Joint centering	$h = \frac{1}{2} \sum_i u_i^2$	Stay near the centres of the joint ranges.
Joint-limit barrier	$h = -\sum_i \log(1 - u_i^2)$	Increasingly penalize approach to a physical stop, for $ u_i < 1$.
Reference posture	$h = \frac{1}{2} \ \mathbf{q} - \mathbf{q}_{\text{ref}}\ ^2$	Remain close to the selected starting posture.

Predict with the Jacobian, correct the pose, then check the step

- 1 Select one legal starting IK from the map and choose the null-space objective and its gain.
- 2 Compute the required tool displacement and the current Jacobian. Add the projected secondary motion to the pseudoinverse prediction.
- 3 Correct the finite-step position and orientation error. A first-order null prediction alone drifts away from a curved self-motion manifold.
- 4 Validate joint limits, full-Jacobian rank, pose accuracy, and continuity along the step. Stop and report the first failed check.

$$\mathbf{q}_{k+1}^{\text{pred}} = \mathbf{q}_k + J_k^+ \Delta \mathbf{x}_k - \gamma \Delta s N_k \nabla h(\mathbf{q}_k)$$

Here $\Delta \mathbf{x}_k = [\Delta \mathbf{p}_k; \text{Log}(R_{k+1} R_k^T)^\vee]$ is a small geometric pose increment, with both components expressed in the world frame.

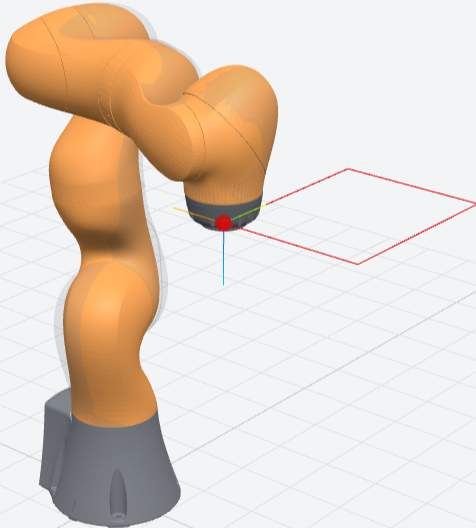
Follow the same rectangle with different starting IKs and costs

Select a starting point at $s = 0$, choose the null-space cost, and build the numerical path. The default rectangle includes a starting solution that completes the path within the joint limits.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path

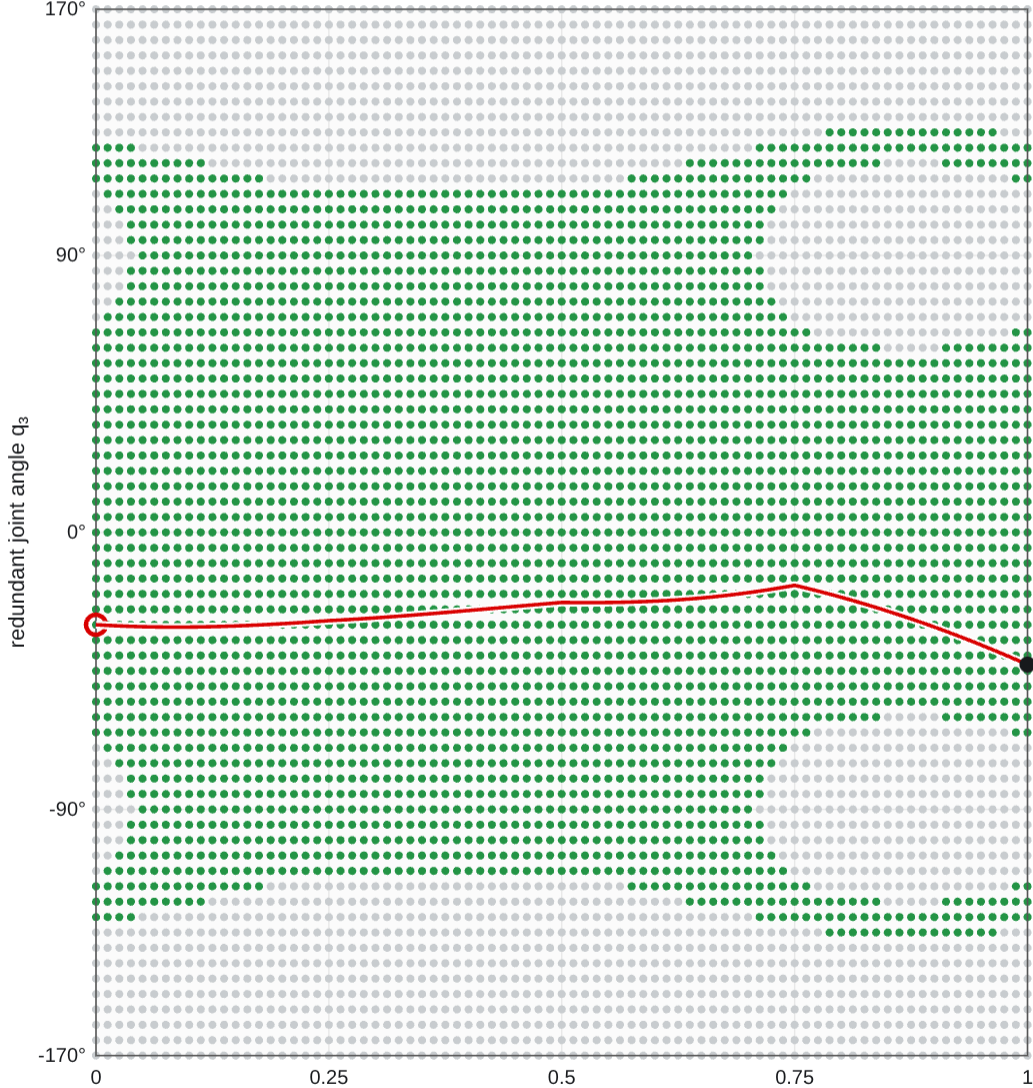


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Path progress × redundant angle

IK 5 · click a green start at $s = 0$



● legal, regular

● limit / singular / no root

● plan on this IK chart

x0.45y0.00z0.60width0.22height0.28m

Rebuild map

Default

Starting redundant angle

-30°

IK 1

limit / rank

IK 2

limit / rank

IK 3

limit / rank

IK 4

legal start

IK 5

legal start

IK 6

limit / rank

IK 7

limit / rank

IK 8

limit / rank

Planner

Numerical · null space

Null-space objective

Joint centering

Null-space gain γ

0.80

Build path

Compare 8 starts

Replay

Reset

IK 5 · numerical / centering

$s = 1.000 \cdot q_3 = -42.9^\circ$

COMPLETE PATH

minimum limit margin 20.0°

max pose error $2.0e-10$

min $\sigma(\hat{J})$ 0.1417

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

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Fixing the redundant angle turns a continuous family into discrete candidates

$$\mathcal{M}_s = \{\mathbf{q} : f(\mathbf{q}) = {}^0T_{t,d}(s)\}$$

$$\mathcal{S}_s(\lambda) = \mathcal{M}_s \cap \{q_3 = \lambda\}$$

Before fixing q_3

A reachable regular pose has a locally one-dimensional inverse image.

After fixing $q_3 = \lambda$

The iiwa model generically has up to eight isolated analytical IK roots. A particular pose and angle can have fewer roots, and some roots violate joint limits.

Solve the arm geometry, then the wrist orientation

1 · Two elbow choices

Let $\mathbf{w}$ run from the shoulder to the wrist centre. The two arm lengths are $a = b = 0.4$ m.

$$c_4 = \frac{\|\mathbf{w}\|^2 - a^2 - b^2}{2ab}$$

$$q_4 = \pm \arccos c_4$$

2 · Two shoulder choices

With $q_3 = \lambda$, set $A = a + bc_4$ and $B = -bs_4 \cos \lambda$.

$$w_z = A \cos q_2 - B \sin q_2$$

This gives two generic q_2 roots. Then use the horizontal wrist coordinates to recover q_1 .

3 · Two wrist choices

Remove the arm rotation from the desired tool rotation:

$$W = ({}^0R_4)^{T0} R_{t,d}$$

Resolve this spherical-wrist rotation into two generic (q_5, q_6, q_7) solutions.

$$2 \times 2 \times 2 = 8 \text{ generic fixed-angle branches}$$

The three $\pm$ signs label fixed-angle algebraic roots; they are not global SEW classes for the redundant robot. Each root is checked by forward kinematics and native joint limits before planning its continuation.

Every dot is an IK evaluation at one path position and redundant angle

Horizontal · path progress

$s \in [0, 1]$ moves around the rectangle. The same position and orientation command applies to every candidate.

Vertical · redundant angle

$\lambda = q_3$ selects a fixed-angle IK slice. Inspect each of the eight branches separately or show all branches.

Green and grey dots

Green: a real root passes the implemented pose, joint-limit, and full-rank checks. **Grey:** no such admissible root at that sample.

A row keeps q_3 constant. A curve through the map changes redundancy along the path. Green dots certify local samples; the connections between them must also be checked.

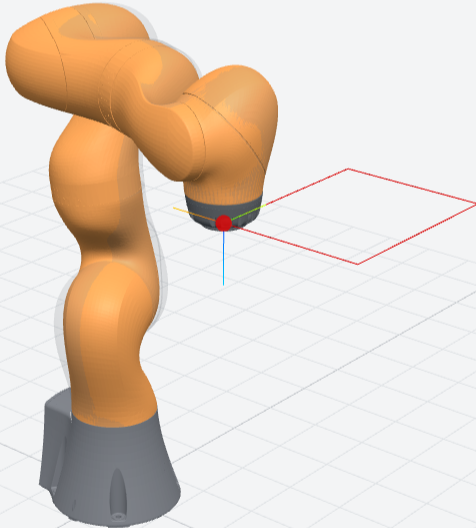
Click a starting IK on the map and compare its possible futures

Green and grey dots come from analytical IK evaluations. Select a legal dot at $s = 0$, then follow it numerically with your chosen null-space cost. Edit the rectangle and rebuild the map to explore a different command.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path

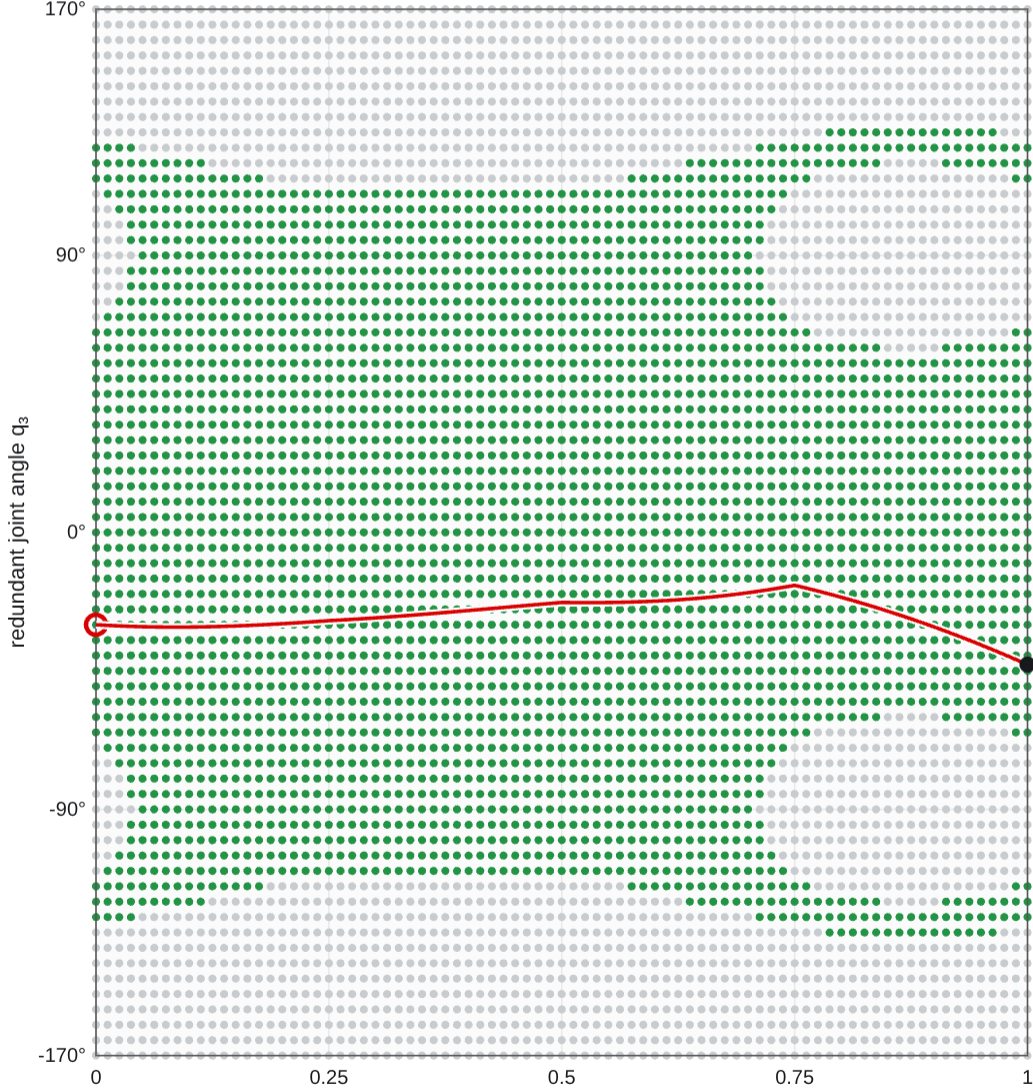


STL opacity

100%

Path progress × redundant angle

IK 5 · click a green start at s = 0



workspace path progress s

● legal, regular ● limit / singular / no root ● plan on this IK chart

Starting redundant angle

-30°

IK 1
limit / rank

IK 2
limit / rank

IK 3
limit / rank

IK 4
legal start

IK 5
legal start

IK 6
limit / rank

IK 7
limit / rank

IK 8
limit / rank

Planner

Numerical · null space

Null-space objective

Joint centering

Null-space gain γ

0.80

Build path

Compare 8 starts

Replay

Reset

IK 5 · numerical / centering

s = 1.000 · q3 = -42.9°

COMPLETE PATH

minimum limit margin 20.0°

max pose error 2.0e-10

min σ(j̇) 0.1417

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

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Search connected IK candidates across the whole rectangle

Nodes

$$\mathbf{q}_k^{(b,r)} \in \mathcal{S}_{s_k}(\lambda_r)$$

At each path layer, enumerate the analytical roots for every sampled redundancy angle.

Edges

Connect nearby configurations in adjacent layers only after checking continuous joint motion, pose tracking, limits, and singularity margin.

Proximity on the 2D plot alone is insufficient.

Whole-path strategy

Search from the selected starting IK to the final layer. Compare joint travel and clearance from joint limits across complete candidate paths.

Analytical planning evaluates the global candidate structure. Numerical continuation resolves motion locally. A failed search on a finite grid does not prove that no continuous path exists.

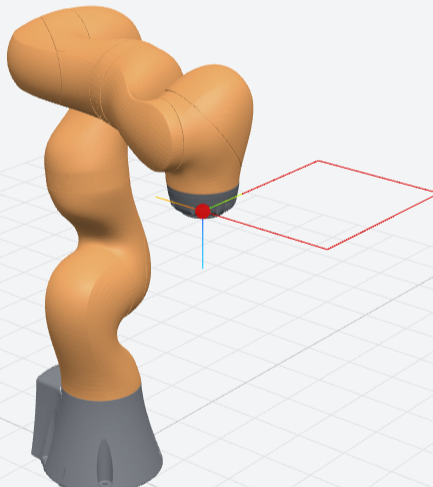
Choose a redundancy schedule using information from the entire path

With the default rectangle and $q_3 = -30^\circ$, try IK 4: numerical joint centering stops, but global planning finds a complete route. Compare the joint-travel and joint-limit objectives, then play their redundancy schedules.

Shared rectangle · fixed tool orientation

KUKA iiwa 7 · native joint limits

black: given path · red: followed path



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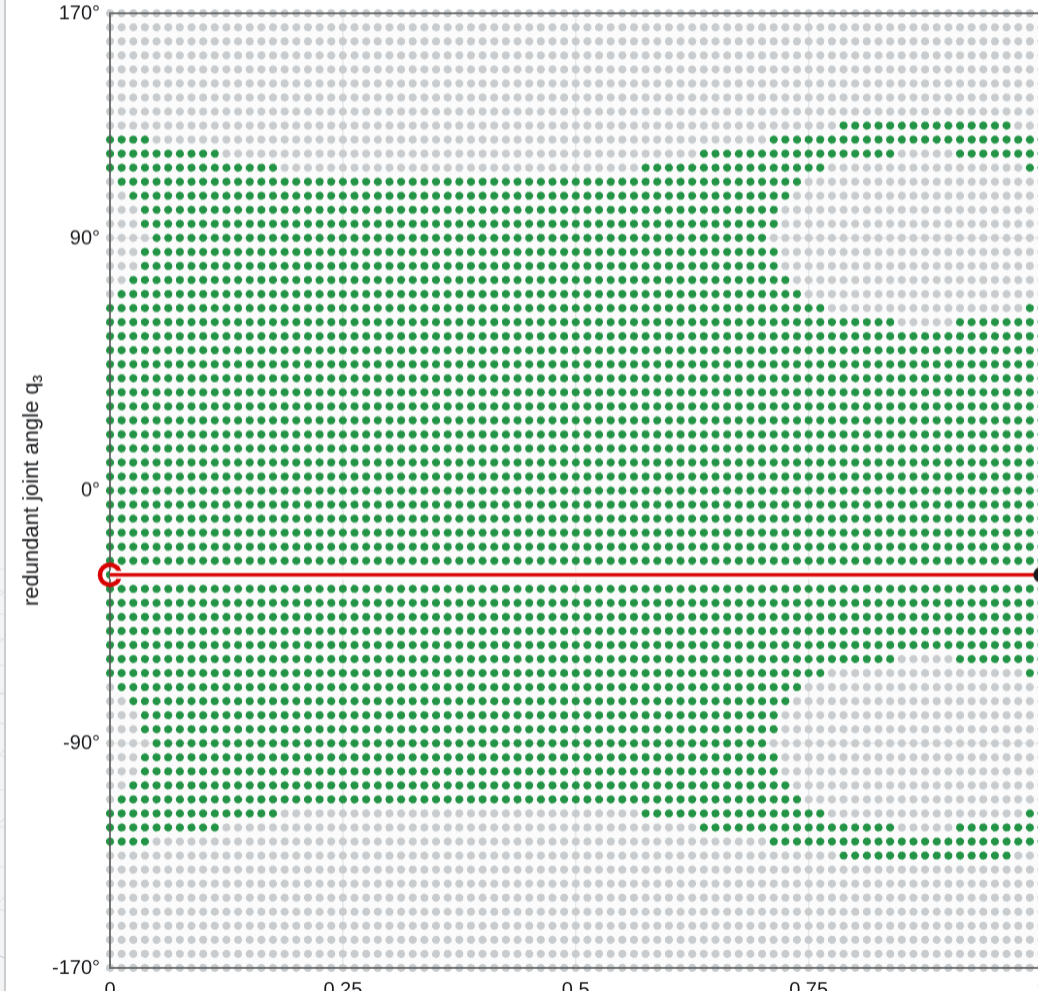
x 0.45y 0.00z 0.60width 0.22height 0.28m

Rebuild mapDefault

Path progress × redundant angle

☐ All 8 maps

IK 5 · click a green start at s = 0



workspace path progress s

● legal, regular

● limit / singular / no root

● plan on this IK chart

Starting redundant angle

-30°

IK 1

limit / rank

IK 2

limit / rank

IK 3

limit / rank

IK 4

legal start

IK 5

legal start

IK 6

limit / rank

IK 7

limit / rank

IK 8

limit / rank

Planner

Analytical · global map

Global edge cost

Least joint travel

Build path

Compare 8 starts

Replay

Reset

IK 5 · analytical / travel

s = 1.000 · $q_3 = -30.0^\circ$

COMPLETE PATH

minimum limit margin 20.0°

max pose error $3.6e-15$

min $\sigma(\dot{J})$ 0.1412

Complete: this start and strategy follow all four edges within the checked limits and singularity margin.

Check the full 6×7 Jacobian before declaring a lost task direction

Fixed-angle chart

$$\bar{J} = J_{[:, 1, 2, 4, 5, 6, 7]} \in \mathbb{R}^{6 \times 6}$$

$\det \bar{J} = 0$ means that holding q_3 fixed prevents local inversion of the remaining six joints. The robot may still have full task rank.

Full redundant robot

$$\text{rank } J < 6 \iff \sigma_{\min}(\hat{J}) = 0$$

The full robot then loses an independent instantaneous task velocity. The redundant coordinate may need to change before this boundary is reached.

Concrete iiwa example: $\mathbf{q} = (20, -35, 30, 50, -40, 0, 60)^\circ$ is within the joint limits. The full scaled Jacobian has $\sigma_{\min} = 0.0477$ m and rank six; removing the q_3 column gives rank five. Here q_3 fails as a local parameter while the tool retains all six instantaneous velocity directions.

Change one choice at a time and inspect the complete joint path

- 1 Keep the rectangle fixed. At one redundant angle, inspect all eight analytical roots and identify the legal starting IKs.
- 2 Keep the start fixed. Compare no secondary cost, joint centering, the joint-limit barrier, and reference-posture attraction.
- 3 Compare numerical continuation with an analytical graph search. Inspect how the chosen redundant angle changes along the path.
- 4 Move or enlarge the rectangle. Rebuild the maps and investigate which starts and strategies still complete the command.

For a closed workspace loop, compare the final and initial joint configurations. Following the same tool loop need not return a redundant arm to the same posture.

Redundancy resolution chooses a continuous route through the IK family

Understand the freedom

The null space identifies instantaneous internal motion that preserves the task velocity.

Choose a local preference

A projected cost gradient selects how numerical planning uses that internal freedom.

Check the global connections

The analytical path–angle maps expose alternative starts and redundancy schedules. Validate the entire connection before accepting a plan.

These demonstrations check kinematic tracking, native joint limits, and singularity margins. Collision checking, dynamics, and a physically smooth time law are additional requirements for execution on hardware.